Fewer Rounds, Fewer Bytes, Smaller Privacy Budgets: Communication-Budgeted Federated Learning for Clinical Prediction over Constrained Links

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***Abstract*—Cross-silo federated learning (FL) between hospitals is limited by the uplink: every round is a scheduled transfer over an institutional WAN, a cellular slice or a satellite link. We treat the job as a network tenant with a communication quota and measure, over 1473 bit-reproducible runs, how the rounds in which a site transmits set accuracy, network cost and the client-level differential-privacy (DP) budget. TierFed solves exactly the integer programme that minimises a local-SGD drift bound under the quota and keeps a per-site head private. At $\rho = 0.25$ it reaches 0.6269 AUROC against 0.6215 for full-rate FedPer, non-inferior at a 0.01 margin with 70% less upload, and 1.6 MB per site with sparse ternary compression. Under DP the transmission schedule is itself observable, so schedulers that react to client updates, ours included, leak information the accountant does not charge. TierFed-DP plans the period and noise multiplier before training from the privacy target, sampling rate and byte quota. At matched budgets $\varepsilon \leq 8$ with Poisson sampling, however, client-level DP reduces every method to near-chance AUROC ($\leq$0.5300); fewer releases help, a fixed period is best, and the planner with its default prior is not better than FedPer. On four real heart-disease institutions TierFed matches full-rate training at a third of the upload, and we check the network model against paced-TCP emulation of a 1,740,290-parameter model (agreement within 2.1%). A fixed period matches or exceeds TierFed everywhere, tier differentiation is rarely active, attacks do not separate the methods, and an earlier negative claim is retracted.**

***Index Terms*—Federated learning, communication efficiency, network resource allocation, 6G edge intelligence, differential privacy, gradient inversion, clinical prediction, non-IID data.**

I. Introduction

FEDERATED learning lets several institutions train one predictive model without exchanging records [1], [2]. In the cross-silo setting that matters for healthcare, the participants are a handful of sites, each behind its own firewall and each connected to the aggregation server by a link that the institution's network team controls [3], [4]. The learning job is then one tenant among many on an uplink with a fixed capacity, a scheduled transfer window and an audit trail. Whether that link is an institutional WAN, a 5G/6G edge slice serving a mobile clinic, or a satellite backhaul serving a remote site, the operator asks how many bytes the job will push per window and whether it will fit [5]–[7].

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Code, per-seed logs and the scripts that generate every table of this article are available at https://github.com/Abraheem13/tierfed.

Prior work reduces communication either by shrinking each message (quantisation, sparsification, sketching [8], [9]) or by sending messages less often (local steps, adaptive periods, lazy uploads [10]–[12]), and a separate line keeps part of the model local for personalisation [13]–[17]. Layer-wise schedules that synchronise different parameter blocks at different frequencies [18] sit between these lines. What is less studied is that the number of transmitting rounds is not only a network quantity. Under client-level differential privacy (DP) [19], every transmitting round is a release that the privacy accountant must charge for and into which noise is injected. A communication quota therefore also sets a privacy cost, and the two can be traded jointly.

This article makes that coupling explicit and measures it. We cast the federated job as a tenant with a quota ρ on the mean fraction of the shared model transmitted per round, and study how accuracy, network cost and DP budget respond when rounds are spent sparingly. We propose TierFed, a scheduler that measures a per-tier dispersion proxy online and solves, exactly, the integer programme that minimises the drift term of a blockwise local-SGD bound subject to ρ, while keeping a per-site classifier head private. We compare it with full-rate personalisation (FedPer), drift correction (FedProx, SCAFFOLD), layer-wise adaptive intervals (a FedLAMA-style reimplementation), adaptive-period local SGD (AdaComm), lazy uploads (LAG) and fixed multi-rate schedules, all on one shared architecture table, with ten seeds, Holm-corrected paired tests and one-sided non-inferiority on both AUROC and AUPRC.

Findings. On a patient-deduplicated readmission benchmark:

- *Synchronising less does not cost accuracy.* Every sparse schedule we evaluated, including fixed periods and TierFed, matches or slightly exceeds full-rate FedPer; TierFed at $\rho = 0.25$ is non-inferior to FedPer on AUROC and AUPRC with 70% less upload (Section VIII-A). An earlier version of this work reported the opposite for fixed schedules; that result came from patient leakage across splits and a scheduling bug, and we retract it.
- *Timing and size reductions compose.* With sparse ternary compression and error feedback, TierFed uploads 1.6 MB per site instead of FedPer's 217.8 MB without loss of AUROC (Section VIII-C).
- *Schedules are releases.* Under client-level DP the rounds

in which a site transmits are observable, so adaptive schedulers (including adaptive TierFed, AdaComm, LAG and FedLAMA) leak through their schedule. A data-independent planner, TierFed-DP, chooses the period and the noise multiplier jointly from the privacy target, the sampling rate and the byte quota, and is compared with full-rate and fixed-period methods at matched (ε, δ) with subsampling amplification. The result is negative: at $\varepsilon \leq 8$ over 20 sites every method is near chance, and the planner does not beat a sparse fixed period (Sections IV-F, VIII-F).

- *Real sites and real transport.* The comparison is repeated on a federation of four real institutions, and the analytic network costs are checked against trace-driven emulation over paced TCP with a model of 1,740,290 parameters (Sections VIII-H, VIII-E).

We are equally explicit about the limits: a well-chosen fixed period and AdaComm reach comparable or better accuracy at comparable upload without DP; TierFed's divergence-driven differentiation between tiers is rarely active on this data; and neither update inversion nor membership inference separates the methods. The contribution is therefore an operator-facing scheduler with an exact budget solver, together with a careful, reproducible measurement of how the communication quota simultaneously sets network cost and privacy cost, rather than an accuracy improvement.

Contributions.

- A network-side formulation of cross-silo FL as a tenant with a communication quota, and a blockwise local-SGD bound whose drift term $\sum_g K_g^2 \tau_{\max}^2 (\Gamma_g + \sigma_g^2)$ motivates per-tier intervals (Sections III, V).
- TierFed: an exact, sub-millisecond solver for the budget-constrained interval programme, driven by online dispersion measurements, composable with any compressor (Section IV).
- The observation that data-dependent schedules are not covered by the standard client-level DP accountant, a data-independent privacy- and byte-budgeted planner (TierFed-DP), and a matched-ε comparison with Poisson sampling and the subsampled-Gaussian accountant (Sections VI-D, IV-F, VIII-F).
- Network-level costing on five link profiles, validated by socket-level emulation, real-institution benchmarks, and an attack-surface evaluation on the slices each method actually transmits (Sections VIII-D–VIII-H).
- A reporting discipline: bit-identical reruns, patient-grouped splits, a single architecture table, Holm-corrected tests, non-inferiority on AUROC and AUPRC, every number generated from released logs, and an explicit list of configurations in which the method does not help.

II. Related Work

Aggregation under heterogeneity. FedAvg [1] averages all parameters every round. FedProx [20] adds a proximal term; SCAFFOLD [21] corrects client drift with control variates at twice the payload. Surveys catalogue many variants [2], [22]; all synchronise everything every round.

Personalisation. FedPer [13] keeps a private classification head; FedRep [14] alternates representation and head updates; LG-FedAvg [15] keeps lower layers local and shares the top; FedBABU [16] trains only the body during federation; Pillutla *et al.* [17] give a general treatment of partial-model personalisation; Ditto [23] and FedBN [24] personalise through regularisation and normalisation statistics. FedPer is the strongest baseline in our experiments and is the reference for every non-inferiority test. None of these methods controls *when* the shared part is sent.

Communication frequency. Local SGD with a fixed period is analysed in [10], [25]. AdaComm [11] adapts a single period from the training loss to minimise an error-versus-runtime bound; LAG [12] lets a client skip an upload when its update has changed little since the last one; FedPAQ [26] combines periodic averaging with quantisation. These are the closest prior work to a measurement-driven schedule, and we implement AdaComm and LAG as baselines rather than discussing them away. They differ from TierFed in three respects: they use one period for the whole model (AdaComm) or a per-client rather than per-block decision (LAG), they do not accept an explicit budget, and they do not withhold a personalised head.

Layer-wise aggregation. FedLAMA [18] assigns per-layer intervals from measured discrepancy; pFedLA [27] learns per-layer weights with a hypernetwork; L-DAWA [28] weights layers by divergence in self-supervised FL; FedCDA [29] aggregates across rounds. TierFed differs in casting the schedule as a budget-constrained minimisation of the bound term $\sum_g K_g^2 \Gamma_g$, in exposing ρ as the single operator-facing knob, and in excluding the head from aggregation. We report a reimplementation of FedLAMA from its published description and label it as such.

Compression. Quantisation, sparsification and sketching reduce message size [8], [9]. They are orthogonal to message timing; Section VIII-C shows the two compose multiplicatively.

FL over wireless and 6G networks. The communication cost of FL over wireless links has been studied through joint learning-and-communication design [7], over-the-air computation [30], and surveys of FL in mobile edge networks [5], [6]; 6G roadmaps place distributed learning among native network services [31], [32]. Our contribution is complementary: we do not design the physical layer, we give the application a knob that maps directly onto a slice or window quota and we cost every method on published link profiles.

Privacy and attack surface. Client-level DP for FL follows McMahan *et al.* [19] with Rényi accounting [33]. Gradient and update inversion [34], [35] and membership inference [36] measure what an adversary recovers. We evaluate all three on every method, on the slices each method actually transmits.

III. System Model and Problem Statement

A. Federation and objective

N sites hold private datasets $\mathcal{D}_k$ of size n_k, $n = \sum_k n_k$, $p_k = n_k/n$. The shared objective is $F(\theta) = \sum_k p_k F_k(\theta)$ with $F_k(\theta) = \mathbb{E}_{\xi \sim \mathcal{D}_k}[f(\theta; \xi)]$. In round r each participating

TABLE I
NOTATION.

N, n_k, p_k	sites; local sample count; aggregation weight
θ, θ_g, w_g	shared parameters; tier g; its share of the shared payload
θ_f	private classifier head (never transmitted)
K_g, $K_{\max}$, τ_g	interval of tier g; cap; round of its last synchronisation
Γ_g, $\hat{\Gamma}_g$	gradient divergence of tier g (A3); measured dispersion proxy
ρ	communication quota (mean payload fraction after warm-up)
T_w, R, τ_k	warm-up rounds; rounds; local steps per round at site k
η, L, σ_g^2	step size; smoothness; per-tier gradient variance
C, σ, (ε, δ)	DP clipping norm; noise multiplier; privacy budget
δ_{NI}	non-inferiority margin

TABLE II
LINK PROFILES USED BY THE NETWORK COST MODEL (3). VALUES ARE REPRESENTATIVE TARGETS, NOT MEASUREMENTS, AND ARE THE ONLY ASSUMPTIONS IN THE NETWORK-LEVEL RESULTS.

Profile	β (Mbit/s)	ℓ (ms)	P (W)	W (s)
NR cell edge [37]	5	30	0.2	60
NR mid-cell [37]	50	20	0.2	60
6G edge slice (target) [32]	1000	5	0.5	60
LEO satellite uplink	10	40	2.0	60
Institutional WAN (wired)	100	10	0	60

site initialises the shared part of its model from the broadcast state and performs one local epoch of $\tau_k = \lceil n_k/B \rceil$ optimiser steps with batch size B (Adam in the main experiments; an SGD arm is in the supplement), then returns the shared slice that is due. Table I collects the notation.

B. Tier partition

Parameters are partitioned into disjoint tiers $\mathcal{G} = \{\mathrm{s}, \mathrm{m}, \mathrm{f}\}$ (slow, medium, fast). Membership is structural: parameters are ordered by their position in the forward computation graph and cut at depth fractions $\varphi_1 < \varphi_2$, with the classifier head always assigned to the fast tier. For the architectures in this article the model declares the cut explicitly (input projection, hidden blocks, head), which the structural rule reproduces. The fast tier is private, which recovers FedPer-style personalisation as the special case $K_{\mathrm{s}} = K_{\mathrm{m}} = 1$. The parameter counts and tier shares of every model are taken from the model instantiated on the real data and reported in Section VII, Table IV.

C. Multi-rate aggregation and payload

Let $\mathcal{A}_r \subseteq \mathcal{G}$ be the tiers due in round r and $\mathcal{S}_r$ the participating sites. The server aggregates only the active tiers,

$$\theta_g^{(r+1)} = \begin{cases} \sum_{k \in \mathcal{S}_r} n_k \, \theta_{k,g}^{(r)} \big/ \sum_{k \in \mathcal{S}_r} n_k, & g \in \mathcal{A}_r, \\ \theta_g^{(r)}, & g \notin \mathcal{A}_r, \end{cases} \tag{1}$$

and sites keep their own fast tier throughout. With w_g the share of the shared payload carried by tier g, the mean payload fraction over R rounds is

$$\Pi(\{K_g\}) = \frac{1}{R} \sum_{r=1}^{R} \sum_{g \in \mathcal{A}_r} w_g \;\longrightarrow\; \sum_{g \in \mathcal{G}_{\mathrm{sh}}} \frac{w_g}{K_g}. \tag{2}$$

Rounds with $\mathcal{A}_r = \emptyset$ transmit nothing; these local-only rounds are the mechanism by which a quota below $\min_g w_g$ is realised. The realised fraction differs from ρ because warm-up rounds are always full and the head is never sent; both the target and the realised fraction are reported.

D. Network cost model

The learning job is costed on a link profile (β, ℓ, P): uplink rate β (bit/s), round-trip latency ℓ, transmit power P. Sites upload in parallel on independent links, so a round in which b_k bits are sent by site k costs

$$t_r = \max_k \frac{b_k}{\beta} + \ell, \qquad E_r = \sum_k \frac{b_k}{\beta} P, \tag{3}$$

and nothing when no site sends. Communication time and energy are summed over rounds; window compliance is the fraction of rounds whose t_r fits a scheduled window W. Table II lists the five profiles, chosen to span a cell-edge 5G NR uplink, a mid-cell NR uplink, a 6G sub-THz edge slice target, an LEO satellite uplink and a shared institutional WAN. The model deliberately omits scheduling, fading and retransmission; it is a cost translation of measured bits under stated assumptions, and every method is costed on the same profiles.

E. Problem statement

Given a quota $\rho \in (0, 1]$ on the mean shared-payload fraction, choose the synchronisation intervals $\{K_g\}$ each round so that the federated model's accuracy is non-inferior to the full-rate personalised baseline while the payload satisfies (2) and the attack surface, measured as in Section VI, does not grow.

IV. TIERFED: A BUDGET-OPTIMAL CONTROLLER

A. Divergence measurement

TierFed estimates the heterogeneity of each tier from the dispersion of the slices it actually receives. For tier g with tensors $j \in g$,

$$\hat{\Gamma}_g^{(r)} = \frac{\sum_{j \in g} \frac{1}{|\mathcal{S}_r|} \sum_{k \in \mathcal{S}_r} \left\| \theta_{k,j}^{(r)} - \bar{\theta}_j^{(r)} \right\|_2^2}{\sum_{j \in g} \left\| \bar{\theta}_j^{(r)} \right\|_2^2 + \epsilon}, \tag{4}$$

a scale-free ratio that makes tiers of different magnitude comparable. Two properties of this estimator are stated now so that no claim rests on them implicitly. First, it is available only for tiers received in round r; for the others the last estimate is carried forward and its age is logged. Second, because the received slices are the result of $K_g \tau_k$ local steps, $\hat{\Gamma}_g$ scales with the interval that produced it (Section V); it is a proxy for the drift the bound penalises, not an unbiased estimate of Γ_g.

Algorithm 1 TierFed: server round r

Require: quota ρ, warm-up T_w, cap $K_{\max}$, intervals $\{K_g\}$, last-sync $\{\tau_g\}$
determine $\mathcal{A}_r$ by (7)
if $\mathcal{A}_r = \emptyset$ **then**
  return {local-only round: zero payload}
end if
broadcast $\{\theta_g^{(r)}\}_{g\in\mathcal{A}_r}$ to $\mathcal{S}_r$
for all $k \in \mathcal{S}_r$ in parallel **do**
  load broadcast tiers; restore private head $\theta_{f,k}$
  run one local epoch; return (optionally compressed) active tiers
end for
(if DP) clip each site's delta to C, sum, add $\mathcal{N}(0, \sigma^2 C^2 I)$, divide by $|\mathcal{S}_r|$
aggregate active tiers by (1)
update $\hat{\Gamma}_g$ for $g \in \mathcal{A}_r$ by (4); $\tau_g \leftarrow r$
$\{K_g\} \leftarrow$ exact solution of (5)

B. The integer programme

Given a quota ρ, the controller solves

$$\{K_g^\star\} = \arg\min_{K_g\in\{1,\dots,K_{\max}\}} \sum_{g\in\mathcal{G}_{\rm sh}} K_g^2\,\hat{\Gamma}_g \quad \text{s.t.} \quad \sum_{g\in\mathcal{G}_{\rm sh}} \frac{w_g}{K_g} \le \rho. \tag{5}$$

With at most three shared tiers and $K_{\max} = 8$ the feasible set has at most $8^3 = 512$ points, so (5) is solved exactly by enumeration in well under a millisecond (Table XXIV). We also report the greedy doubling rule of an earlier version of this work, which lengthens the interval of the tier with the smallest marginal bound cost per unit of bandwidth saved,

$$g^\star = \arg\min_g \frac{\hat{\Gamma}_g\big[(2K_g)^2 - K_g^2\big]}{w_g/K_g - w_g/(2K_g)} = \arg\min_g \frac{6\,\hat{\Gamma}_g K_g^3}{w_g}, \tag{6}$$

and which reaches only powers of two. The closed form exposes the controller's preference: sparsity goes first to tiers with low divergence, a large payload share and a short current interval.

C. Scheduling rule and warm-up

Tier g is due when $r - \tau_g \ge K_g^\star$, where τ_g is the round of its last synchronisation:

$$\mathcal{A}_r = \begin{cases} \mathcal{G}_{\rm sh}, & r \le T_w, \\ \{g:\ r - \tau_g \ge K_g^\star\}, & r > T_w. \end{cases} \tag{7}$$

Divergence is largest in the earliest rounds, so the first T_w rounds synchronise every shared tier. $T_w = 5$ and $K_{\max} = 8$ are held fixed across every dataset, architecture, loss and privacy setting in this article; a T_w sweep is in Table XX.

D. Composition with compression

TierFed controls *when* a tier is sent; a compressor controls *how large* the message is. For a compressor with rate $c \in (0, 1]$ bits per parameter relative to float32 the expected upload over R rounds is

$$B = R\,|\mathcal{S}|\,\Pi(\{K_g\})\,c\,32\,d_{\rm sh} \text{ bits}, \tag{8}$$

with $d_{\rm sh}$ the shared parameter count; the reduction factors multiply. Section VIII-C tests this on five compressors.

E. What the operator sets

The only operator-facing quantity is ρ. It is a quota on the fraction of the shared model transmitted per round on average, so it maps directly onto a slice allocation or a transfer-window budget through (8) and (3). The paper reports the mapping $\rho \mapsto$ (upload, communication time, energy) on every link profile so that an operator can read off what a given quota buys.

F. Planning under a privacy budget (TierFed-DP)

Under client-level DP the set of rounds in which a site transmits is visible to the aggregator and to an eavesdropper. A schedule computed from client updates, as in adaptive TierFed, AdaComm, LAG or FedLAMA, is therefore itself a data-dependent release that the Gaussian accountant does not cover (Section VI-D). TierFed-DP removes this channel by planning the schedule *before* training from public quantities only: the target $(\varepsilon^\star, \delta)$, the sampling rate q, the horizon R, the warm-up T_w, the tier shares w_g and the quota ρ. For every candidate $K \in \{1,\dots,K_{\max}\}^{|\mathcal{G}|}$ it computes the release set $\mathcal{R}(K) = \{r \le R:\ \exists g,\ r \le T_w \text{ or } (r - T_w) \bmod K_g = 0\}$, the smallest noise multiplier $\sigma(K)$ for which $|\mathcal{R}(K)|$ releases of the subsampled Gaussian mechanism are $(\varepsilon^\star, \delta)$-DP, and the surrogate

$$J(K) = \sum_g w_g K_g^2 \;+\; \lambda\,\frac{1}{R}\sum_{r\in\mathcal{R}(K)} \Big(\sum_{g \text{ due at } r} w_g\Big)\,\sigma(K)^2, \tag{9}$$

the drift term of (11) with a flat, data-free heterogeneity prior plus the noise energy injected into the shared model per round. Fewer releases need less noise per release, since σ^2 grows roughly linearly with the release count at fixed $\varepsilon^\star$, but drift more; (9) trades the two, subject to the byte quota $\frac{1}{R}\sum_r \sum_{g \text{ due}} w_g \le \rho$. The minimiser is found by enumeration ($K_{\max}^{|\mathcal{G}|} = 64$ candidates) and the server is told $\sigma(K)$. The single prior λ absorbs the unknown smoothness and heterogeneity constants of the bound; we report its sensitivity rather than tune it. Because γ is flat, TierFed-DP chooses equal intervals for both shared tiers; it is a privacy-aware *period* planner, not a per-tier one, and we say so in the results.

V. Convergence Analysis

The analysis is for the shared backbone with the private heads frozen at their current values, i.e. $F_k(\theta) = F_k(\theta; \theta_{f,k})$; the personalisation gap this introduces is not bounded and is measured empirically by the FedPer-versus-FedAvg contrast of Table VIII. We make the standard assumptions of local-SGD analysis [10], [25], stated per tier.

Assumption 1 (A1–A4)**.** *Each F_k is L-smooth; stochastic gradients are unbiased with per-tier variance $\mathbb{E}\|\nabla_g f(\theta;\xi) - \nabla_g F_k(\theta)\|^2 \le \sigma_g^2$; the per-tier heterogeneity is bounded,*

$$\frac{1}{N}\sum_{k=1}^{N} \big\|\nabla_g F_k(\theta) - \nabla_g F(\theta)\big\|^2 \le \Gamma_g \quad \forall\theta,\ \forall g; \tag{10}$$

and $F \geq F^\star$.

Proposition 1 (Multi-rate local-SGD bound)**.** *Under A1–A4, with local SGD of step size η, $\tau_k \in [\tau_{\min}, \tau_{\max}]$ local steps per round, and tier g averaged every K_g rounds, there are absolute constants c_0, c_1, c_2 such that for $\eta \leq c_0/(LK_{\max}\tau_{\max})$,*

$$\frac{1}{R}\sum_{r=1}^{R}\mathbb{E}\big\|\nabla F(\bar{\theta}_r)\big\|^2 \leq \underbrace{\frac{c_1\,(F(\bar{\theta}_1)-F^\star)}{\eta\,\tau_{\min}R}}_{\textit{optimisation}} + \underbrace{c_1\,\eta L\sum_g \frac{\sigma_g^2}{N}}_{\textit{stochastic}} + \underbrace{c_2\,\eta^2L^2\sum_g K_g^2\tau_{\max}^2\big(\Gamma_g+\sigma_g^2\big)}_{\textit{multi-rate drift}}. \tag{11}$$

Proof sketch. Apply the perturbed-iterate argument of [25] blockwise. Between two synchronisations of tier g a site takes at most $K_g\tau_{\max}$ local steps on that block, so the squared deviation of its block iterate from the virtual average is bounded by $\eta^2K_g^2\tau_{\max}^2(\Gamma_g+\sigma_g^2)$ up to the constants of the cited theorem; the blocks are disjoint, so the deviations add, and the descent lemma with the step-size condition absorbs the cross terms. Setting $K_g = 1$ for a single block recovers the local-SGD rate with τ local steps. The derivation at the level of the cited theorem is in the supplement. □

The design rule. The cost of a long interval is $K_g^2\tau_{\max}^2(\Gamma_g+\sigma_g^2)$: quadratic in the interval, linear in heterogeneity. A tier tolerates sparsity if and only if its divergence is small. Because Γ_g depends on the data, the architecture and the training stage, the bound gives no reason to expect one fixed schedule to be right in general; whether this matters in practice is an empirical question, and on the benchmark of Section VIII-A it largely does not.

What the controller observes. The controller cannot observe Γ_g. It observes the dispersion (4) of the slices it receives, which, by the same argument as in the proof, is itself of order $\eta^2K_g^2\tau^2(\Gamma_g+\sigma_g^2)$ after K_g rounds of local training. $\hat{\Gamma}_g$ is therefore a noisy, scale-free proxy for the drift term of tier g at its current interval. We log $\hat{\Gamma}_g$ together with the interval that produced it and we do not claim that $\hat{\Gamma}_g$ estimates Γ_g.

Optimiser. The bound is for SGD; the benchmark uses Adam with persistent state for every method because that is the solver the baselines were introduced with and that practitioners use. An SGD arm is reported in Table XI; it shows that the ordering of methods does depend on the local solver.

What is not claimed. Nothing above proves that (5) is the right programme, only that its objective is the shape of an upper bound on the drift term. The controller is evaluated empirically against fixed schedules, FedLAMA-style adaptive intervals, AdaComm and LAG.

VI. Threat Model and Privacy Mechanisms

A. Threat model

We consider an honest-but-curious aggregator and a passive eavesdropper on the uplink, both of whom observe exactly the bytes a site transmits in each round: the broadcast state and the site's returned slice of the active tiers. Neither party observes the private head, local data or local optimiser state. We do not consider Byzantine participants or model poisoning. The attack surface of a method is therefore the sequence of slices it transmits, and we measure it in three ways on every method.

B. Update inversion

The adversary reconstructs a batch $\hat{x}$ from an observed update $\Delta_{\mathcal{A}}$ restricted to the transmitted tiers by solving

$$\hat{x} = \arg\min_x \; 1 - \cos\Big(\nabla_{\theta_{\mathcal{A}}}\ell\big(h_\theta(x), y\big),\; -\Delta_{\mathcal{A}}\Big), \tag{12}$$

the cosine objective of Geiping *et al.* [35] without an image prior, since the inputs are tabular. Unlike a single-batch gradient attack [34], the observed update is the result of a full local epoch of Adam steps, which is the update an aggregator actually sees; this is the update-inversion setting of [35]. The attack is run inside training, on the first site's real transmitted slice, in every round in which that site transmits, for every method, so a method that transmits rarely is attacked on the rounds in which it does. Success is the normalised reconstruction error and the feature correlation between each true row and its best-matching reconstruction; lower correlation means a smaller attack surface. The attack is intentionally generic: the aim is a like-for-like comparison of what each method exposes, not the strongest possible adversary.

C. Membership inference

Against the deployed (personalised) model of every site we run the loss-threshold attack of Yeom *et al.* [36]: the adversary labels a record a training member when the model's loss on it is low. The attack AUC over members and patient-grouped non-members, averaged over sites, measures memorisation of the released model; 0.5 is no leakage.

D. Client-level differential privacy

Each round, every site joins the cohort $\mathcal{S}_r$ independently with probability q (Poisson sampling; $q = 1$ is full participation). The server clips every participating site's returned delta and adds noise to the sum before dividing by the *fixed* expected cohort size:

$$\tilde{\Delta} = \frac{1}{qN}\Big(\sum_{k\in\mathcal{S}_r}\Delta_k\min\big(1, \tfrac{C}{\|\Delta_k\|_2}\big) + \mathcal{N}(0,\sigma^2C^2I)\Big), \tag{13}$$

the fixed-denominator DP-FedAvg estimator of McMahan *et al.* [19], so that adding or removing one site changes the sum by at most C in ℓ_2 and the denominator reveals nothing about the realised cohort. A planned release whose cohort happens to be empty still adds noise. The mechanism is applied by the server to every method identically. The accountant composes, over the R_{tx} planned releases, the Rényi DP of the Poisson-subsampled Gaussian mechanism at integer orders [38],

$$\varepsilon_\alpha = \frac{R_{\text{tx}}}{\alpha-1}\log\sum_{k=0}^{\alpha}\binom{\alpha}{k}(1-q)^{\alpha-k}q^k e^{\frac{k^2-k}{2\sigma^2}}, \qquad \varepsilon = \min_\alpha\Big\{\varepsilon_\alpha + \frac{\log(1/\delta)}{\alpha-1}\Big\} \tag{14}$$

with $\delta = 10^{-5}$; for $q = 1$ this reduces to $R_{\text{tx}}\alpha/(2\sigma^2)$ [33]. The implementation reproduces the reference value of the

TensorFlow-Privacy MNIST tutorial ($q = 256/60000$, $\sigma = 1.1$, 60 epochs, $\varepsilon \approx 3.0$) and the $q = 1$ closed form; both are unit tests. For a matched-privacy comparison every method is given the same target $\varepsilon^\star$ and σ is calibrated, by bisection on (14), to that method's own release count.

Which schedules are covered. The accountant covers a release only if the decision to release is independent of the private data. This holds for FedAvg, FedProx, FedPer, the fixed period and TierFed-DP, whose release rounds are fixed before training. It does not hold for adaptive TierFed, AdaComm, LAG or FedLAMA: their schedules are functions of client updates or losses, so the release pattern leaks information that (14) does not charge for. The released code refuses to run these strategies under DP (an explicit override exists only to reproduce the earlier table). The earlier version of this article compared all methods at equal σ including the adaptive ones; that comparison is formally not covered by its stated budgets, is moved to the Supplement (Section S3) with this caveat, and is superseded by the matched-ε comparison in Section VIII-F. Earlier still, the aggregate rather than each site's delta was clipped; that mechanism does not bound the sensitivity to one site. The budgets reported are as computed; with $N = 20$ sites, client-level budgets in the range we test are weak guarantees, and we claim only that the trade-off is measured correctly.

VII. Experimental Design

A. Data

Diabetes 130-US Hospitals [39] is the main tabular benchmark: 30-day readmission from encounter-level features. Cleaning follows Strack *et al.*: encounters discharged to death or hospice are removed and only the first encounter per patient is kept, which leaves 55,992 training and 13,998 test encounters from 55,992 patients, 160 encoded features, and a positive rate of 9.0%. The public release carries no hospital identifier, so sites are formed from the medical-specialty column by longest-processing-time bucketing with a client-share cap of 15%: a specialty larger than the cap, including the "unknown specialty" group that covers roughly half of the encounters, is split across clients, patient-grouped, before bucketing. The largest client therefore holds 12% of the training data and the mean label total-variation distance between clients is 0.009. We call these clients *sites* rather than hospitals. The train/test split and every per-client held-out split are patient-grouped.

Real institutions. Fed-Heart-Disease [40], [41] pools the four UCI Heart Disease collections (Cleveland Clinic, Hungarian Institute of Cardiology, University Hospitals Zürich and Basel, VA Long Beach) with the target "any angiographic disease"; it has 4 clients and 736 training patients, and missingness indicators are added because two features are mostly missing outside Cleveland. The open **eICU-CRD demo** [42], [43] was also prepared, one client per `hospitalid`, but no two of its hospitals have 80 or more stays, so it was not used. The heart federation is small; it tests whether the conclusions survive real site heterogeneity, not clinical performance. The full credentialed eICU-CRD is supported by the same loader and was not run. **PneumoniaMNIST** [44] is the imaging benchmark: a natively binary 28×28 collection, partitioned by Dirichlet label skew ($\alpha = 0.3$) with a minimum client size of 50 images. Table III summarises; all dataset statistics in this article are read from the run logs.

TABLE III
Dataset characteristics, from the run logs.

	Diabetes 130-US	PneumoniaMNIST
Modality	tabular EHR	imaging
Train / test rows	55,992 / 13,998	5,232 / 624
Patients (train)	55,992	–
Features / channels	160	1 (28×28)
Positive rate (%)	9.0	74.2
Sites	20 (specialty, capped)	20 (Dirichlet 0.3)
Largest site share (%)	12	14
Label TV (mean)	0.009	0.370

TABLE IV
Automatic tier partition on the real data (from the run logs). The fast tier is the private head.

Architecture	Params	Slow (%)	Medium (%)	Fast (%)
MLP (Diabetes)	45,506	45.3	54.4	0.3
CNN (PneumoniaMNIST)	287,586	3.4	96.5	0.1

B. Models and tiers

Four backbones share one architecture table used by every method: an MLP (hidden 128–128–64), a tabular ResNet (width 128, depth 6), an FT-Transformer (width 64, depth 3, 8 heads) [45] and a small CNN for imaging. A wide MLP (hidden 1024–1024–512, 1,740,290 parameters on Diabetes) is used for the link-emulation experiment, so that each upload is several megabytes. Table IV reports the parameter counts and tier shares of each model as instantiated on the real data.

C. Protocol

$R = 60$ rounds, 20 sites at full participation, one local epoch, batch 64, Adam at 10^{-3} with persistent state, inverse-frequency class weighting, $T_w = 5$, $K_{\max} = 8$, $\rho = 0.25$ unless stated. Baselines use their published defaults (FedProx $\mu = 0.01$; FedLAMA-style $\tau = 5$, $\phi = 2$; AdaComm $\tau_0 = 8$; LAG $\xi = 1$); none were tuned on the test data and neither was TierFed. Because personalised methods hold private heads, every method is evaluated on each site's patient-grouped held-out split with that site's own parameters and the per-site metrics are sample-weighted; sites whose held-out split contains a single class cannot yield an AUROC and are excluded identically for every method, with their count reported. A pooled AUROC over all held-out predictions and the global model's AUROC on the untouched test set are reported alongside. Compressors act on each site's update relative to the broadcast with per-client error feedback, and communication is counted from what the compressor emitted for the sites that actually sent. In the fixed multi-rate schedule, as in TierFed, a round in which no tier is due is local-only.

Differential privacy. The matched-privacy comparison uses $C = 1$, $\delta = 10^{-5}$, targets $\varepsilon^\star \in \{1, 2, 4, 8\}$ at Poisson rate

$q = 0.5$ and $\varepsilon^\star = 4$ at $q = 1$, and the four DP-eligible methods FedAvg, FedPer, a fixed period $(K_s, K_m) = (4, 4)$ with $T_w = 5$, and TierFed-DP with $\lambda = 1$ ($\lambda \in \{0.1, 10\}$ as sensitivity). Each method's σ is calibrated to its own planned release count; final-round AUROC is reported, since best-round selection on held-out data is not covered by the budget.

Link emulation. The analytic cost model (3) is checked against real transport. Each run's logged per-site upload sizes are replayed, round by round, as concurrent loopback TCP connections, one per transmitting site, whose senders are paced by a token bucket at the profile's uplink rate after a one-way delay of half the RTT; the round ends when the last byte arrives. This is application-layer, trace-driven emulation: it includes TCP segmentation, socket buffering and operating-system scheduling, but not radio scheduling, fading, loss or the downlink, and kernel-level emulation (netem) was not available on the test machine. Time-to-target adds the measured compute time of the slowest site per round.

D. Determinism

Repeated runs of one configuration are bit-identical on the machine that produced the logs (deterministic kernels, single-threaded reductions, seeded loaders); the reproduction script asserts this before any run. An earlier version of this work lacked this guarantee and identical configurations differed by up to 0.006 AUROC, which is larger than the effect it tested.

E. Statistics

All comparisons are paired across the ten shared seeds. We report Holm-corrected paired t-test p-values over the family of all method-versus-FedPer contrasts, BCa bootstrap intervals of the paired difference, and the paired standardised difference d_z with the small-sample correction. Parity is never inferred from a non-significant difference. A one-sided non-inferiority test at margin $\delta_{\mathrm{NI}} = 0.01$ is run on *both* final AUROC and final AUPRC,

$$t_{\mathrm{NI}} = \frac{\bar{d} + \delta_{\mathrm{NI}}}{s_d/\sqrt{n}}, \qquad p = 1 - \mathcal{T}_{n-1}(t_{\mathrm{NI}}), \tag{15}$$

and the non-inferiority tests are reported uncorrected and interpreted against the Holm-corrected difference tests. The margin is a reporting convention: 0.01 AUROC is below the between-seed standard deviation of every method in Table V; it has no clinical derivation and we do not claim one. Early-stopped and last-ten-round statistics are reported so that no conclusion depends on the final round alone.

VIII. Results

Every table body and every number in this section is generated from the per-seed logs by the released analysis script. Unless stated otherwise, values are means over ten seeds on Diabetes 130-US with the MLP backbone; p-values are from paired tests across seeds.

A. Main benchmark

Table V reports the main benchmark. Full-rate FedPer reaches 0.6215 AUROC at 217.8 MB per site. TierFed at $\rho = 0.25$ reaches 0.6269 (paired difference +0.0054, $p = 0.006$) at 65.3 MB, transmitting in 18 of 60 rounds; it is non-inferior to FedPer on AUROC ($p =< 10^{-4}$) and on AUPRC ($p =< 10^{-4}$). The round-sparse baselines are at least as accurate: AdaComm reaches 0.6302 at 29.1 MB (TierFed minus AdaComm: -0.0033, $p = 0.024$), and the fixed multi-rate schedule ($K_{\mathrm{s}} = 5$, $K_{\mathrm{m}} = 2$) reaches 0.6339 at 82.9 MB. LAG at its default threshold never skips an upload and is equivalent to full-rate averaging (0.6351); SCAFFOLD with the Adam local solver used for every method underperforms (0.5930) and is reported with its native SGD solver in Table XI. Early-stopped AUROC is within a narrow band for all methods, so no conclusion depends on the final round.

Synchronising less does not hurt here. Table VII sweeps the fixed schedule's periods. A uniform period of four rounds for both shared tiers reaches 0.6407 at 58.1 MB, the best accuracy of any configuration in this study; only a long slow-tier period ($K_{\mathrm{s}} = 8$) costs accuracy (0.6130 with $K_{\mathrm{m}} = 1$). Table VIII separates the private head from the schedule: FedPer and FedAvg are level (0.6215 and 0.6203), so the head does not matter on this data, and TierFed with and without the head is also level (0.6269 and 0.6264). The modest gain of every sparse schedule over full-rate averaging is consistent with additional local training acting as personalisation. An earlier version of this work reported that fixed multi-rate schedules lose about 0.05 AUROC to FedPer; that finding was produced by encounters of the same patient appearing in training and test splits, and by a fallback rule that sent the medium tier whenever no tier was due. Both are corrected here and the finding does not reproduce.

B. The budget knob and the controller

Table IX shows that ρ behaves as an operator knob: upload rises monotonically with the quota, from 45.7 MB at $\rho = 0.15$ to 167.2 MB at $\rho = 0.8$. Accuracy is flat above $\rho = 0.25$ and drops only at $\rho = 0.15$ (0.6070); the best quota is $\rho = 0.4$ (0.6366 at 84.9 MB). The last column is the honest test of the divergence-driven mechanism: at $\rho = 0.25$ the solver gave the two shared tiers different intervals in 0% of rounds, at $\rho = 0.4$ in 7%, and at most in 26% at $\rho = 0.15$. On this data the controller mostly reduces to a uniform period chosen to meet the quota, which explains why a well-chosen fixed period performs as well. Where the solvers differ, exactness helps: at $\rho = 0.4$ the exact solver reaches 0.6366 at 84.9 MB against 0.6283 at 95.7 MB for the greedy doubling rule (Table X).

Table XI gives the baselines a fair frontier. AdaComm with $\tau_0 = 8$ is the most byte-efficient configuration without DP (0.6302 at 29.1 MB); LAG skips uploads only at $\xi = 10^4$, where its accuracy collapses (0.5673). With an SGD local solver SCAFFOLD recovers (0.6383), FedAvg and FedPer improve (0.6396, 0.6379), and TierFed is lower (0.6251); the ordering of methods therefore depends on the local solver, which we state as a limitation.

TABLE V
MAIN BENCHMARK (DIABETES 130-US, 20 SITES, 60 ROUNDS, 10 SEEDS). FINAL-ROUND FEDERATED AUROC (MEAN ± SD), AUPRC, EARLY-STOPPED AUROC, POOLED AUROC, CUMULATIVE UPLOAD PER SITE, NUMBER OF TRANSMITTING ROUNDS, AND ONE-SIDED NON-INFERIORITY p-VALUES AGAINST FEDPER AT $\delta_{\mathrm{NI}} = 0.01$ ON AUROC AND AUPRC.

Method	n	AUROC	AUPRC	ES-AUROC	Pooled	Upload (MB)	Tx rounds	$\mathrm{NI}_{\mathrm{AUROC}}$ p	$\mathrm{NI}_{\mathrm{AUPRC}}$ p
FedAvg	10	0.6203 ± 0.0100	0.171	0.6417	0.6209	218.4	60	$< 10^{-4}$	$< 10^{-4}$
FedProx	10	0.6279 ± 0.0107	0.175	0.6433	0.6290	218.4	60	$< 10^{-4}$	$< 10^{-4}$
SCAFFOLD	10	0.5930 ± 0.0137	0.138	0.6119	0.5930	436.9	60	0.997	1.000
FedPer	10	0.6215 ± 0.0087	0.171	0.6419	0.6201	217.8	60	(ref.)	(ref.)
FedLAMA	10	0.6291 ± 0.0101	0.176	0.6421	0.6295	154.1	60	$< 10^{-4}$	$< 10^{-4}$
AdaComm	10	0.6302 ± 0.0071	0.168	0.6365	0.6290	29.1	8	$< 10^{-4}$	< 0.001
LAG	10	0.6351 ± 0.0124	0.179	0.6407	0.6362	218.4	60	$< 10^{-4}$	$< 10^{-4}$
Fixed schedule	10	0.6339 ± 0.0100	0.177	0.6395	0.6338	82.9	37	$< 10^{-4}$	$< 10^{-4}$
TierFed	10	0.6269 ± 0.0082	0.175	0.6415	0.6261	65.3	18	$< 10^{-4}$	$< 10^{-4}$

TABLE VI
HOLM-CORRECTED PAIRED COMPARISONS ON FINAL AUROC AGAINST FEDPER (10 SEEDS): MEAN DIFFERENCE, BCA 95% INTERVAL, CORRECTED p, PAIRED d_z.

Comparison	Δ	95% CI	p_{Holm}	d_z	Sig.
AdaComm vs FedPer	+0.0087	[0.0070, 0.0105]	< 0.001	2.66	✓
FedAvg vs FedPer	-0.0012	[-0.0038, 0.0004]	0.578	-0.33	
FedLAMA vs FedPer	+0.0076	[0.0058, 0.0101]	0.001	1.91	✓
FedProx vs FedPer	+0.0065	[0.0042, 0.0085]	0.005	1.57	✓
Fixed schedule vs FedPer	+0.0124	[0.0101, 0.0156]	< 0.001	2.44	✓
LAG vs FedPer	+0.0136	[0.0100, 0.0166]	< 0.001	2.23	✓
SCAFFOLD vs FedPer	-0.0284	[-0.0397, -0.0196]	0.005	-1.57	✓
TierFed vs FedPer	+0.0054	[0.0023, 0.0079]	0.036	1.03	✓

TABLE VII
FIXED-SCHEDULE PERIOD SWEEP $(K_{\mathrm{s}}, K_{\mathrm{m}})$, 10 SEEDS; ROUNDS WITH NO TIER DUE ARE LOCAL-ONLY.

$(K_{\mathrm{s}}, K_{\mathrm{m}})$	n	AUROC	Upload (MB)
(2, 1)	10	0.6317 ± 0.0097	170.0
(2, 2)	10	0.6362 ± 0.0110	112.5
(4, 1)	10	0.6314 ± 0.0102	145.3
(4, 2)	10	0.6370 ± 0.0112	87.8
(4, 4)	10	0.6407 ± 0.0114	58.1
(8, 1)	10	0.6130 ± 0.0079	132.1
(8, 2)	10	0.6155 ± 0.0077	74.6
(8, 4)	10	0.6254 ± 0.0073	44.9
(8, 8)	10	0.6250 ± 0.0088	29.0

TABLE VIII
ATTRIBUTION (10 SEEDS): PRIVATE HEAD × SCHEDULE, AND TIERFED WITH AND WITHOUT THE PRIVATE HEAD.

Cell	n	AUROC	AUPRC	Upload (MB)
fedavg	10	0.6203 ± 0.0100	0.171	218.4
fedper	10	0.6215 ± 0.0087	0.171	217.8
fixed	10	0.6339 ± 0.0100	0.177	82.9
sched_only	10	0.6336 ± 0.0102	0.177	83.5
tierfed	10	0.6269 ± 0.0082	0.175	65.3
tierfed_nohead	10	0.6264 ± 0.0073	0.174	65.5

TABLE IX
QUOTA SWEEP FOR TIERFED (10 SEEDS): ACCURACY, UPLOAD, REALISED PAYLOAD FRACTION, MEAN INTERVALS CHOSEN BY THE EXACT SOLVER, AND THE SHARE OF POST-WARM-UP ROUNDS IN WHICH THE TWO SHARED TIERS RECEIVED DIFFERENT INTERVALS.

ρ	n	AUROC	AUPRC	Upload (MB)	Realised	$\bar{K}_{\mathrm{s}}$	$\bar{K}_{\mathrm{m}}$	Differentiated (%)
0.15	10	0.6070 ± 0.0119	0.163	45.7	0.21	–	–	–
0.25	10	0.6269 ± 0.0082	0.175	65.3	0.30	–	–	–
0.40	10	0.6366 ± 0.0119	0.179	84.9	0.39	–	–	–
0.60	10	0.6290 ± 0.0093	0.176	116.2	0.53	–	–	–
0.80	10	0.6296 ± 0.0094	0.176	167.2	0.77	–	–	–

C. Scheduling and compression compose

Table XII confirms (8): reductions from timing and from message size multiply. 8-bit quantisation alone takes FedPer from 217.8 to 54.5 MB at unchanged accuracy (0.6213); TierFed with 8-bit quantisation uploads 16.4 MB (0.6266); and TierFed with sparse ternary compression uploads 1.6 MB at 0.6290, against 5.2 MB (0.6250) for FedPer with the same compressor. Count-sketch compression fails for every method (0.4986, 0.5017, 0.5049): at the sketch width used, the unsketched estimate of the update is too noisy for this model, and we report it rather than tune it away.

D. Network-level cost

Table XIII translates measured bits into what an operator sees. On a cell-edge NR uplink the total communication time of the job falls from 19.2 s for FedPer to 5.8 s for TierFed. The model is small, so every method fits the 60-s window on every link profile; the absolute times matter less than the proportional reductions, which carry over to larger models with the same schedule.

E. A wide model on emulated links

With 1,740,290 parameters, a full-rate upload is several megabytes per site and the link dominates the round on every constrained profile (Table XV). The analytic model agrees with real paced TCP to within 2.1% on every run, so the network costs reported in Table XIII are not an artefact of the formula; the residual is the RTT term, which the analytic model charges in full and the emulator as a one-way delay, plus transport overhead. Over 60 rounds on cell-edge NR, the emulated upload time is 669 s for FedPer, 201 s for TierFed, 178 s for the fixed period and 100 s for AdaComm. Time to

TABLE X
EXACT SOLVER VERSUS GREEDY DOUBLING AT TWO QUOTAS (10 SEEDS).

ρ / solver	n	AUROC	Upload (MB)
0.15 / exact	10	0.6070 ± 0.0119	45.7
0.25 / exact	10	0.6269 ± 0.0082	65.3
0.25 / greedy	10	0.6269 ± 0.0082	65.3
0.40 / exact	10	0.6366 ± 0.0119	84.9
0.40 / greedy	10	0.6283 ± 0.0112	95.7
0.60 / exact	10	0.6290 ± 0.0093	116.2
0.80 / exact	10	0.6296 ± 0.0094	167.2

TABLE XI
BASELINE SWEEPS (10 SEEDS): ADACOMM INITIAL PERIOD τ_0, LAG THRESHOLD ξ, AND THE SGD LOCAL SOLVER FOR FEDAVG, FEDPER, SCAFFOLD AND TIERFED.

Configuration	n	AUROC	AUPRC	Upload (MB)	Tx rounds
adacomm tau16	10	0.6125 ± 0.0111	0.152	18.2	5
adacomm tau2	10	0.6301 ± 0.0095	0.177	109.2	30
adacomm tau4	10	0.6275 ± 0.0068	0.174	54.6	15
adacomm tau8	10	0.6302 ± 0.0071	0.168	29.1	8
fedavg sgd	10	0.6396 ± 0.0109	0.172	218.4	60
fedper sgd	10	0.6379 ± 0.0110	0.170	217.8	60
lag xi10	10	0.6351 ± 0.0124	0.179	218.4	60
lag xi100	10	0.6351 ± 0.0124	0.179	218.4	60
lag xi1000	10	0.6352 ± 0.0126	0.179	168.4	46
lag xi10000	10	0.5673 ± 0.0498	0.135	6.8	2
scaffold sgd	10	0.6383 ± 0.0106	0.173	436.9	120
tierfed sgd	10	0.6251 ± 0.0095	0.157	65.3	18

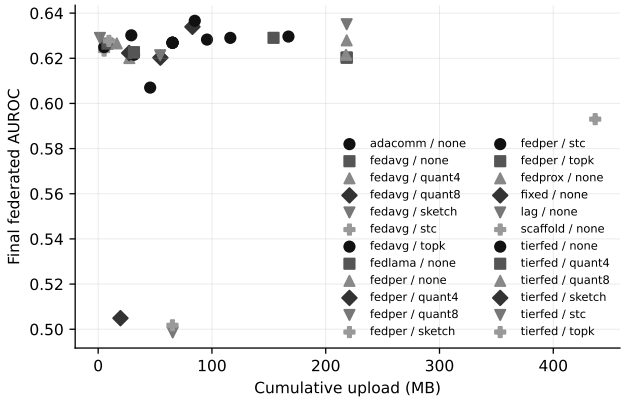


Fig. 1. Accuracy–communication frontier on Diabetes (10 seeds): every method, quota and compressor on one pair of axes.

TABLE XII
SCHEDULING × COMPRESSION (10 SEEDS). COMPRESSORS ACT ON THE UPDATE WITH PER-CLIENT ERROR FEEDBACK. TIERFED ROWS USE $\rho = 0.25$.

Method / compressor	n	AUROC	AUPRC	Upload (MB)
FedAvg / none	10	0.6203 ± 0.0100	0.171	218.4
FedAvg / quant4	10	0.6201 ± 0.0094	0.172	27.4
FedAvg / quant8	10	0.6203 ± 0.0101	0.171	54.7
FedAvg / sketch	10	0.4986 ± 0.0075	0.090	65.5
FedAvg / stc	10	0.6234 ± 0.0113	0.172	5.2
FedAvg / topk	10	0.6216 ± 0.0107	0.171	31.5
FedPer / none	10	0.6215 ± 0.0087	0.171	217.8
FedPer / quant4	10	0.6223 ± 0.0097	0.171	27.3
FedPer / quant8	10	0.6213 ± 0.0085	0.172	54.5
FedPer / sketch	10	0.5017 ± 0.0043	0.091	65.3
FedPer / stc	10	0.6250 ± 0.0099	0.173	5.2
FedPer / topk	10	0.6227 ± 0.0092	0.172	31.4
TierFed / none	10	0.6269 ± 0.0082	0.175	65.3
TierFed / quant4	10	0.6265 ± 0.0091	0.174	8.2
TierFed / quant8	10	0.6266 ± 0.0083	0.174	16.4
TierFed / sketch	10	0.5049 ± 0.0058	0.092	19.6
TierFed / stc	10	0.6290 ± 0.0087	0.174	1.6
TierFed / topk	10	0.6277 ± 0.0087	0.174	9.4

a low target (AUROC 0.60) is not reduced by TierFed (23 s against 23 s), because the target is reached during the full-rate warm-up; the saving is in the remaining rounds. AdaComm never reaches 0.62 within 60 rounds. On accuracy (Table XIV) the wide model favours sparse synchronisation more strongly than the small one: the fixed period reaches 0.6352, AdaComm 0.6134 and TierFed 0.5999, against 0.5843 for FedPer and 0.5802 for FedAvg; with three seeds this is indicative, and the fixed period is again the most accurate.

F. Attack surface and differential privacy

Table XVI compares methods at equal privacy rather than equal noise, and the result is negative for every method. At budgets $\varepsilon^\star \leq 8$ with 20 sites, the noise required for client-level DP reduces every method to near-chance AUROC (between 0.5073 and 0.5300 at $q = 0.5$); none of these models would be usable. Within that range the ordering is the one the release count predicts: a full-rate method makes 60 releases and needs $\sigma = 5.18$ at $\varepsilon^\star = 4$, $q = 0.5$; the fixed period makes 19 releases at $\sigma = 3.09$ and is the most accurate method at every budget, significantly better than FedPer at $\varepsilon^\star = 8$ (+0.014, $p = 0.030$) and at $q = 1$, $\varepsilon^\star = 4$ (+0.012, $p = 0.018$). TierFed-DP with its default prior $\lambda = 1$ is level with FedPer (+0.000, $p = 0.961$ at $\varepsilon^\star = 4$) and below the fixed period at every budget, though not significantly (-0.008, $p = 0.202$): at $q = 0.5$ its surrogate (9) overweights drift and selects $K = 2$ with 32 releases, more than the fixed period. Table XVII confirms the mechanism: raising the noise weight to $\lambda = 10$ selects $K = 7$ and 12 releases and gives the highest AUROC of the planner. The planner therefore responds to λ as designed, but its default prior is mis-calibrated for this regime, and we do not claim that it improves on choosing the sparsest fixed period. Subsampling amplification works as expected on the accounting side, lowering σ at equal ε (FedPer 5.18 at $q = 0.5$ against 10.04 at $q = 1$), without changing the conclusion. The adaptive schedulers are absent by construction, since their release patterns depend on client data; the earlier equal-σ comparison that included them is in the Supplement (Section S3) with that caveat.

Without DP, neither attack separates the methods (Table XVIII). Update-inversion feature correlation is low for every method (TierFed 0.089, FedPer 0.088, AdaComm 0.065), and membership-inference AUC is close to chance (TierFed 0.540, FedPer 0.535). Per transmitted round, TierFed exposes the same slice as FedPer; its reduction in attack surface is only that it transmits in fewer rounds, which this generic attack does not exploit.

TABLE XIII
NETWORK COST OF THE MAIN BENCHMARK UNDER THE LINK PROFILES OF TABLE II: TOTAL COMMUNICATION TIME ON CELL-EDGE NR, LEO SATELLITE AND THE 6G EDGE-SLICE TARGET, UPLINK ENERGY ON CELL-EDGE NR, AND THE FRACTION OF ROUNDS FITTING A 60-S WINDOW ON CELL-EDGE NR. MEANS OVER 10 SEEDS, COMPUTED FROM MEASURED BITS BY (3).

Method	Upload (MB)	t NR-edge (s)	t LEO (s)	t 6G (s)	E NR-edge (J)	Window (%)
FedAvg	218.4	19	11	0.4	69.9	100
FedProx	218.4	19	11	0.4	69.9	100
SCAFFOLD	436.9	37	20	0.5	139.8	100
FedPer	217.8	19	11	0.4	69.7	100
FedLAMA	154.1	14	9	0.4	49.3	100
AdaComm	29.1	3	1	0.1	9.3	100
LAG	218.4	19	11	0.4	69.9	100
Fixed schedule	82.9	8	5	0.2	26.5	100
TierFed	65.3	6	3	0.1	20.9	100

TABLE XIV
WIDE MLP (1,740,290 PARAMETERS) ON DIABETES (3 SEEDS): ACCURACY AND UPLOAD.

Method	n	AUROC	AUPRC	Upload (MB)	Params
AdaComm	3	0.6134 ± 0.0033	0.159	1253.0	1,740,290
FedAvg	3	0.5802 ± 0.0095	0.143	8353.4	1,740,290
FedPer	3	0.5843 ± 0.0058	0.144	8348.5	1,740,290
Fixed schedule	3	0.6352 ± 0.0091	0.175	2226.3	1,740,290
TierFed	3	0.5999 ± 0.0047	0.156	2504.5	1,740,290

G. Robustness, architectures and imaging

TierFed matches or exceeds FedPer under every class-imbalance loss (Table XIX; e.g. LDAM 0.6276 versus 0.6150). A longer warm-up buys a small gain at a proportional cost in bytes (Table XX; $T_w = 20$: 0.6354 at 108.9 MB). On the six-block tabular ResNet, TierFed reaches 0.6279 at 603.5 MB against 0.6137 at 2006.0 MB for FedPer, and on the FT-Transformer 0.6345 at 165.2 MB against 0.6352 at 581.2 MB (Table XXI). As on the MLP, the fixed multi-rate schedule is at least as accurate as TierFed on every backbone (ResNet 0.6382 at 1006.8 MB; FT-Transformer 0.6386 at 270.7 MB), while TierFed uploads less. The FT-Transformer rows other than the fixed schedule were trained on GPU with the fused attention kernel, which is not bit-deterministic; they are reported over five seeds with that caveat.

On PneumoniaMNIST (Table XXII) the picture differs from the tabular benchmark and we report it as it is. The two methods that share the classifier head are the most accurate, FedAvg (0.9959) and the FedLAMA-style schedule (0.9965); TierFed reaches 0.9698 at 364.1 MB against 0.9643 at 1213.7 MB for FedPer, and is not non-inferior to FedPer at the 0.01 margin ($p = 0.182$). Under strong label skew the private head is a liability rather than an asset: both head-private methods trail the head-sharing ones and FedPer has a large between-seed spread. A schedule that keeps the head private should therefore not be used for this kind of task without a shared head.

H. Real institutions

On the four heart-disease institutions, TierFed reaches 0.8131 AUROC at 8.50 MB, significantly above FedPer (0.8016 at 28.32 MB; difference +0.012, $p = 0.009$) and level with FedAvg (+0.000, $p = 0.967$) and the fixed period (0.8171 at 7.55 MB; $p = 0.221$); SCAFFOLD is the only method that is clearly worse. The ordering of the specialty-bucket benchmark therefore survives real institutional heterogeneity: sparse schedules match full-rate training at about a third of the upload, and the fixed period is as good as TierFed. The federation is small (736 training patients), so between-seed variance is large; it is not a clinical result. DP was not run on it: with four clients the client-level Gaussian mechanism needs noise of the order of the whole aggregate.

I. Scalability

Table XXIV reports measured wall-clock and accuracy from 10 to 100 sites with the Dirichlet partition, since the specialty partition cannot produce 100 sites. Upload per site scales linearly with the number of sites for every method and TierFed keeps its proportional saving at every size; seconds per round are dominated by local training and are similar across methods. Accuracy at 100 sites is lower for TierFed than for FedAvg, consistent with each site holding less data. The controller's exact solve enumerates at most 512 points and takes well under a millisecond at every size.

IX. DISCUSSION

What the evidence supports. Three claims are supported by ten-seed paired tests on patient-deduplicated data. First, on this benchmark synchronising less does not cost accuracy; every sparse schedule matches or slightly exceeds full-rate personalisation, and TierFed is non-inferior to FedPer on AUROC and AUPRC at a fraction of the upload. Second, timing and message-size reductions compose multiplicatively when compressors act on the update with error feedback, giving more than two orders of magnitude less upload at unchanged accuracy. Third, under client-level DP the release count is a design variable that must be fixed before training: a schedule that reacts to client updates is itself a leak, and once the schedule is planned, the period trades drift against the noise each release must carry. At matched $\varepsilon \leq 8$ over 20 sites, however, no method retains useful utility, so on this benchmark client-level DP is not yet practical at any communication rate; fewer releases help at the margin, and the sparsest fixed period is the best choice. A communication quota and a privacy budget therefore have to be negotiated together, which matters operationally because they are usually set by different stakeholders.

What it does not support. TierFed is not more accurate than simpler round-sparse alternatives. A fixed uniform period of four rounds is the most accurate configuration we ran, and AdaComm reaches comparable accuracy with fewer bytes when DP is off. The divergence-driven mechanism that motivates per-tier intervals rarely activates on this data; at most quotas the exact solver returns the same interval for both shared tiers. Under DP, neither adaptive TierFed nor AdaComm is eligible at all, and TierFed-DP with a flat prior always chooses equal intervals for both tiers, so it is a privacy-aware period planner rather than a per-tier scheduler. The case for TierFed over these alternatives is operational rather than statistical: it exposes a quota that maps directly onto a slice or window allocation, solves for the schedule exactly,

TABLE XV
TRACE-DRIVEN SOCKET EMULATION OF THE WIDE-MLP RUNS: TOTAL EMULATED UPLOAD TIME (REAL TCP OVER PACED LOOPBACK LINKS), THE ANALYTIC MODEL'S PREDICTION FROM (3), THEIR RELATIVE DIFFERENCE, TOTAL COMPUTE TIME OF THE SLOWEST SITE, AND WALL-CLOCK TIME TO REACH AUROC 0.60 AND 0.62 ("NEVER" IF NOT REACHED IN 60 ROUNDS). MEANS OVER SEEDS; THE CELL-EDGE PROFILE WAS EMULATED FOR ONE SEED.

Profile	Method	n	Emulated (s)	Analytic (s)	Diff. (%)	Compute (s)	$t_{0.60}$ (s)	$t_{0.62}$ (s)
hospital_wan	AdaComm	3	5.2	5.1	-1.7	10	7	never
hospital_wan	FedAvg	3	34.6	34.0	-1.7	10	1	2
hospital_wan	FedPer	3	34.6	34.0	-1.6	10	2	2
hospital_wan	Fixed schedule	3	9.2	9.1	-1.7	10	1	11
hospital_wan	TierFed	3	10.4	10.2	-1.9	10	2	2
nr_typical	AdaComm	3	10.2	10.2	-0.2	10	10	never
nr_typical	FedAvg	3	68.2	68.0	-0.2	10	2	3
nr_typical	FedPer	3	68.1	68.0	-0.2	10	3	4
nr_typical	Fixed schedule	3	18.2	18.1	-0.2	10	1	16
nr_typical	TierFed	3	20.4	20.4	-0.2	10	3	4
nr_cell_edge	AdaComm	1	100.5	100.5	0.1	10	24	never
nr_cell_edge	FedAvg	1	669.6	670.1	0.1	11	13	24
nr_cell_edge	FedPer	1	669.2	669.7	0.1	10	23	34
nr_cell_edge	Fixed schedule	1	178.5	178.6	0.1	10	11	94
nr_cell_edge	TierFed	1	200.8	200.9	0.1	10	23	34

TABLE XVI
MATCHED-PRIVACY COMPARISON (10 SEEDS, $C = 1$, $\delta = 10^{-5}$). EVERY METHOD IS RUN AT THE SAME TARGET $\varepsilon^\star$ WITH POISSON CLIENT SAMPLING AT RATE q; ITS NOISE MULTIPLIER σ IS CALIBRATED TO ITS OWN NUMBER OF PLANNED RELEASES WITH THE SUBSAMPLED-GAUSSIAN ACCOUNTANT (14). ACHIEVED ε, FINAL-ROUND AUROC AND AUPRC, AND UPLOAD. ONLY DATA-INDEPENDENT SCHEDULES ARE ELIGIBLE (SECTION VI-D).

q / $\varepsilon^\star$ / Method	n	σ	Releases	ε	AUROC	AUPRC	Upload (MB)
0.5 / 1 / FedAvg	10	19.14	60	1.00	0.5073 ± 0.0163	0.092	108.7
0.5 / 1 / FedPer	10	19.14	60	1.00	0.5102 ± 0.0092	0.093	108.4
0.5 / 1 / Fixed schedule	10	10.95	19	1.00	0.5150 ± 0.0083	0.095	33.7
0.5 / 1 / TierFed-DP	10	10.68	18	1.00	0.5137 ± 0.0081	0.095	31.8
0.5 / 2 / FedAvg	10	9.84	60	2.00	0.5079 ± 0.0193	0.092	108.7
0.5 / 2 / FedPer	10	9.84	60	2.00	0.5090 ± 0.0092	0.093	108.4
0.5 / 2 / Fixed schedule	10	5.72	19	2.00	0.5178 ± 0.0149	0.098	33.7
0.5 / 2 / TierFed-DP	10	6.25	23	2.00	0.5118 ± 0.0105	0.098	40.6
0.5 / 4 / FedAvg	10	5.18	60	4.00	0.5104 ± 0.0198	0.093	108.7
0.5 / 4 / FedPer	10	5.18	60	4.00	0.5135 ± 0.0089	0.096	108.4
0.5 / 4 / Fixed schedule	10	3.09	19	4.00	0.5218 ± 0.0134	0.108	33.7
0.5 / 4 / TierFed-DP	10	3.88	32	4.00	0.5137 ± 0.0094	0.099	57.6
0.5 / 8 / FedAvg	10	2.84	60	8.00	0.5148 ± 0.0173	0.093	108.7
0.5 / 8 / FedPer	10	2.84	60	8.00	0.5162 ± 0.0107	0.100	108.4
0.5 / 8 / Fixed schedule	10	1.76	19	8.00	0.5300 ± 0.0163	0.116	33.7
0.5 / 8 / TierFed-DP	10	2.16	32	8.00	0.5177 ± 0.0120	0.110	57.6
1.0 / 4 / FedAvg	10	10.04	60	4.00	0.5100 ± 0.0192	0.093	218.4
1.0 / 4 / FedPer	10	10.04	60	4.00	0.5174 ± 0.0103	0.096	217.8
1.0 / 4 / Fixed schedule	10	5.65	19	4.00	0.5294 ± 0.0083	0.113	69.0
1.0 / 4 / TierFed-DP	10	6.22	23	4.00	0.5211 ± 0.0121	0.108	83.5

TABLE XVII
SENSITIVITY OF TIERFED-DP TO ITS PRIOR λ IN (9) AT $q = 1$, $\varepsilon^\star = 4$ (10 SEEDS): CHOSEN INTERVAL, CALIBRATED σ, RELEASES, AUROC, UPLOAD.

λ	n	K	σ	Releases	AUROC	Upload (MB)
0.1	10	2	7.33	32	0.5237 ± 0.0150	116.2
1.0	10	3	6.22	23	0.5211 ± 0.0121	83.5
10.0	10	7	4.49	12	0.5300 ± 0.0141	43.6

TABLE XVIII
ATTACK SURFACE (10 SEEDS): PARAMETERS EXPOSED PER TRANSMITTING ROUND, UPDATE-INVERSION NORMALISED ERROR AND FEATURE CORRELATION ON THE REAL TRANSMITTED SLICES, MEMBERSHIP-INFERENCE AUC ON THE DEPLOYED MODELS (0.5 = NO LEAKAGE), AND UPLOAD.

Method	n	Params exposed	NMSE ↑	Corr. ↓	MIA AUC ↓	Upload (MB)
AdaComm	10	45,506	11.58	0.065	0.534	29.1
FedAvg	10	45,506	9.04	0.103	0.534	218.4
FedPer	10	45,376	9.04	0.088	0.535	217.8
Fixed schedule	10	27,509	11.93	0.048	0.526	82.9
TierFed	10	45,376	10.45	0.089	0.540	65.3

TABLE XIX
LOCAL OBJECTIVE SWEEP, FEDPER VERSUS TIERFED ($\rho = 0.25$), 10 SEEDS.

Loss / method	n	AUROC	AUPRC	Upload
Class-balanced / FedPer	10	0.6232 ± 0.0108	0.173	217.
Class-balanced / TierFed	10	0.6264 ± 0.0071	0.173	65.
Focal / FedPer	10	0.6198 ± 0.0100	0.170	217.
Focal / TierFed	10	0.6237 ± 0.0089	0.173	65.
LDAM / FedPer	10	0.6150 ± 0.0114	0.164	217.
LDAM / TierFed	10	0.6276 ± 0.0091	0.171	65.
Inverse-frequency CE / FedPer	10	0.6215 ± 0.0087	0.171	217.
Inverse-frequency CE / TierFed	10	0.6269 ± 0.0082	0.175	65.

and composes with compression and DP accounting, whereas AdaComm exposes a loss-driven period and a fixed schedule exposes nothing an operator can set from a byte budget.

A retraction. An earlier version of this work reported that fixed multi-rate schedules lose about 0.05 AUROC to personalised baselines. With one encounter per patient, patient-grouped splits and the fallback-tier bug removed, that result does not reproduce. Readers who relied on it should not.

Limitations. (i) The sites of the main tabular benchmark are specialty buckets with a share cap, not hospitals; the real-institution federation is small (four heart-disease centres); the open eICU demo has no two hospitals with 80 or more stays and could not be used at that threshold, and the full credentialed eICU-CRD was not run. (ii) The bound is for SGD and the main benchmark uses Adam; the SGD arm shows that the ordering of methods depends on the local solver. (iii) $\hat{\Gamma}_g$ is a dispersion proxy that scales with the interval that produced it; the controller is a feedback loop on that proxy whose stability we observe but do not prove. (iv) The network model and the socket emulation omit radio scheduling, fading,

TABLE XX
WARM-UP SWEEP FOR TIERFED ($\rho = 0.25$), 10 SEEDS.

T_w	n	AUROC	AUPRC	Upload (MB)
0	10	0.6270 ± 0.0080	0.173	54.5
2	10	0.6323 ± 0.0092	0.176	58.1
5	10	0.6269 ± 0.0082	0.175	65.3
10	10	0.6289 ± 0.0093	0.176	79.9
20	10	0.6354 ± 0.0112	0.179	108.9

TABLE XXI
ARCHITECTURES ON DIABETES (5 SEEDS), EVERY METHOD ON THE SAME ARCHITECTURE WITHIN A ROW GROUP; PARAMETER COUNT FROM THE INSTANTIATED MODEL.

Backbone / method	n	AUROC	AUPRC	Upload (MB)	Params
mlp / FedAvg	5	0.6236 ± 0.0103	0.171	218.4	0
mlp / FedPer	5	0.6240 ± 0.0087	0.172	217.8	0
mlp / Fixed schedule	5	0.6373 ± 0.0117	0.176	82.9	0
mlp / TierFed	5	0.6298 ± 0.0128	0.175	65.3	0
resnet / FedAvg	5	0.6258 ± 0.0192	0.168	2007.3	0
resnet / FedPer	5	0.6137 ± 0.0107	0.161	2006.0	0
resnet / Fixed schedule	5	0.6382 ± 0.0150	0.177	1006.8	0
resnet / TierFed	5	0.6279 ± 0.0130	0.175	603.5	0
transformer / FedAvg	5	0.6401 ± 0.0134	0.177	581.8	0
transformer / FedPer	5	0.6352 ± 0.0115	0.173	581.2	0
transformer / Fixed schedule	5	0.6386 ± 0.0116	0.172	270.7	0
transformer / TierFed	5	0.6345 ± 0.0083	0.171	165.2	0

TABLE XXII
PNEUMONIAMNIST (CNN, 20 SITES, DIRICHLET $\alpha = 0.3$, 10 SEEDS). COLUMNS AS IN TABLE V.

Method	n	AUROC	AUPRC	ES-AUROC	Pooled	Upload (MB)	Tx rounds	$\mathrm{NI}_{\mathrm{AUROC}}$ p	$\mathrm{NI}_{\mathrm{AUPRC}}$ p
FedAvg	10	0.9959 ± 0.0023	0.991	0.9957	0.9724	1214.8	60	0.037	< 0.001
FedPer	10	0.9643 ± 0.0663	0.986	0.9790	0.9948	1213.7	60	(ref.)	(ref.)
FedLAMA	10	0.9965 ± 0.0018	0.995	0.9960	0.9837	877.4	60	0.036	0.005
Fixed schedule	10	0.9642 ± 0.0677	0.989	0.9871	0.9928	614.8	37	$< 10^{-4}$	0.002
TierFed	10	0.9698 ± 0.0238	0.984	0.9866	0.9669	364.1	18	0.182	0.090

TABLE XXIII
FED-HEART-DISEASE, ONE CLIENT PER REAL INSTITUTION (4 INSTITUTIONS, 10 SEEDS, NO DP): AUROC, AUPRC, UPLOAD AND TRANSMITTING ROUNDS.

Method	n	AUROC	AUPRC	Upload (MB)	Tx roun
AdaComm	10	0.7994 ± 0.0397	0.878	4.2	
FedAvg	10	0.8128 ± 0.0356	0.876	28.4	
FedLAMA	10	0.8184 ± 0.0337	0.881	20.4	
FedPer	10	0.8016 ± 0.0305	0.873	28.3	
FedProx	10	0.8136 ± 0.0352	0.877	28.4	
Fixed schedule	10	0.8171 ± 0.0350	0.886	7.6	
LAG	10	0.8102 ± 0.0324	0.876	28.4	
SCAFFOLD	10	0.7533 ± 0.0479	0.832	56.9	
TierFed	10	0.8131 ± 0.0314	0.884	8.5	

loss and the downlink; the emulation is application-layer, not kernel-level, and the profiles are representative targets rather than measurements. (v) The inversion attack is generic; client-level budgets of $\varepsilon \geq 1$ over 20 sites are weak guarantees, and even at those budgets utility is near chance, so DP results need federations with many more sites. (vi) Poisson participation is used only in the DP experiments; the bound (11) is stated for full participation and does not include a sampling term. (vii) The FT-Transformer arm was trained with a non-deterministic GPU attention kernel; all other results are bit-reproducible. (viii) On the imaging benchmark under strong label skew, the methods that share the classifier head are the most accurate and both head-private methods trail them; the results here do not transfer to that setting. (ix) Ten seeds on one public tabular dataset and one imaging collection is a benchmark, not a clinical validation.

Practical guidance. For a tabular model of this size, an operator who needs a byte budget should set ρ from the transfer window through (8) and (3), compose the schedule with 8-bit quantisation or sparse ternary compression with error feedback, and, when client-level DP is required, fix the schedule before training (a fixed period or TierFed-DP), calibrate σ to its release count, and never use a scheduler that reacts to client updates.

X. CONCLUSION

We treated a cross-silo federated job as a network tenant with an explicit communication quota and measured, on bit-reproducible runs over patient-deduplicated data, how the number of transmitting rounds sets accuracy, network cost and privacy cost together. Synchronising less did not cost accuracy; timing and size reductions composed to more than two orders of magnitude less upload; and under client-level DP the schedule itself turned out to be a release, so it must be planned before training; at matched $\varepsilon \leq 8$ over 20 sites every method was near chance, and a privacy- and byte-budgeted planner did not beat a sparse fixed period. On four real institutions sparse schedules matched full-rate training at a third of the upload, and the network costs held over paced TCP to within 2.1%. TierFed turns a byte quota into an exact per-tier schedule and is non-inferior to full-rate personalisation at a fraction of the upload, but it is not more accurate than a well-chosen fixed period or AdaComm, and its divergence-driven differentiation rarely activates on this data. We report these limits, and retract a finding of an earlier version, so that the operational result, that a communication quota and a privacy budget must be planned jointly and before training, rests on evidence a reader can rerun.

ACKNOWLEDGMENT

Generative artificial-intelligence tools were used to draft figure layouts and to refactor analysis code; the author reviewed, edited and verified all scientific content, code, numbers and figures. An earlier version of this work was submitted elsewhere under the title “Nested Federated Learning”; it is superseded by this article, which corrects reporting and code defects of that version and retracts its central negative result (Supplement, Section S1).

DATA AND CODE AVAILABILITY

Diabetes 130-US Hospitals is public (UCI id 296). PneumoniaMNIST is public (MedMNIST v2). UCI Heart Disease is public; the eICU-CRD demo is open on PhysioNet (ODbL); the full eICU-CRD requires credentialed PhysioNet access. Code, per-seed logs, the scripts that generate every table and figure, and the determinism check are at https://github.com/Abraheem13/tierfed.

TABLE XXIV
SCALABILITY (DIABETES, DIRICHLET PARTITION, 30 ROUNDS, 3 SEEDS): ACCURACY, UPLOAD PER SITE, SECONDS PER ROUND AND PEAK GPU MEMORY, MEASURED, FOR 10 TO 100 SITES.

Sites / method	n	AUROC	Upload (MB)	s/round	Peak mem (MB)
10 / FedAvg	3	0.6127 ± 0.0206	54.6	3.07	80
10 / FedLAMA	3	0.6056 ± 0.0117	43.1	3.03	80
10 / FedPer	3	0.5935 ± 0.0072	54.5	3.03	80
10 / TierFed	3	0.6118 ± 0.0100	20.0	3.01	80
20 / FedAvg	3	0.6187 ± 0.0110	109.2	3.12	87
20 / FedLAMA	3	0.6187 ± 0.0183	85.4	3.09	87
20 / FedPer	3	0.6189 ± 0.0235	108.9	3.11	87
20 / TierFed	3	0.6205 ± 0.0242	39.9	3.04	87
50 / FedAvg	3	0.6128 ± 0.0224	273.0	3.40	108
50 / FedLAMA	3	0.6108 ± 0.0203	219.3	3.33	108
50 / FedPer	3	0.6188 ± 0.0240	272.3	3.37	108
50 / TierFed	3	0.6048 ± 0.0170	99.8	3.21	108
100 / FedAvg	3	0.6068 ± 0.0210	546.1	3.99	142
100 / FedLAMA	3	0.6061 ± 0.0230	437.1	3.86	142
100 / FedPer	3	0.5929 ± 0.0255	544.5	3.86	142
100 / TierFed	3	0.5883 ± 0.0157	199.7	3.57	142

ETHICS, FUNDING AND COMPETING INTERESTS

All datasets are de-identified and publicly available under their respective licences; no human-subject data were collected and no ethics approval was required. The author declares no competing interests and no external funding.

REFERENCES

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[2] P. Kairouz, H. B. McMahan *et al.*, "Advances and open problems in federated learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

[3] N. Rieke, J. Hancox, W. Li *et al.*, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, p. 119, 2020.

[4] I. Dayan, H. R. Roth, A. Zhong *et al.*, "Federated learning for predicting clinical outcomes in patients with COVID-19," *Nature Medicine*, vol. 27, pp. 1735–1743, 2021.

[5] W. Y. B. Lim, N. C. Luong, D. T. Hoang, Y. Jiao, Y.-C. Liang, Q. Yang, D. Niyato, and C. Miao, "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 2031–2063, 2020.

[6] S. Niknam, H. S. Dhillon, and J. H. Reed, "Federated learning for wireless communications: Motivation, opportunities, and challenges," *IEEE Communications Magazine*, vol. 58, no. 6, pp. 46–51, 2020.

[7] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Transactions on Wireless Communications*, vol. 20, no. 1, pp. 269–283, 2021.

[8] F. Sattler, S. Wiedemann, K.-R. Müller, and W. Samek, "Robust and communication-efficient federated learning from non-i.i.d. data," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 9, pp. 3400–3413, 2020.

[9] D. Rothchild, A. Panda, E. Ullah, N. Ivkin, I. Stoica, V. Braverman, J. Gonzalez, and R. Arora, "FetchSGD: Communication-efficient federated learning with sketching," in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 8253–8265.

[10] S. U. Stich, "Local SGD converges fast and communicates little," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

[11] J. Wang and G. Joshi, "Adaptive communication strategies to achieve the best error-runtime trade-off in local-update SGD," in *Proc. Machine Learning and Systems (MLSys)*, 2019, pp. 212–229.

[12] T. Chen, G. B. Giannakis, T. Sun, and W. Yin, "LAG: Lazily aggregated gradient for communication-efficient distributed learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018, pp. 5050–5060.

[13] M. G. Arivazhagan, V. Aggarwal, A. K. Singh, and S. Choudhary, "Federated learning with personalization layers," *arXiv preprint arXiv:1912.00818*, 2019.

[14] L. Collins, H. Hassani, A. Mokhtari, and S. Shakkottai, "Exploiting shared representations for personalized federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 2089–2099.

[15] P. P. Liang, T. Liu, L. Ziyin, N. B. Allen, R. P. Auerbach, D. Brent, R. Salakhutdinov, and L.-P. Morency, "Think locally, act globally: Federated learning with local and global representations," *arXiv preprint arXiv:2001.01523*, 2020.

[16] J. Oh, S. Kim, and S.-Y. Yun, "FedBABU: Toward enhanced representation for federated image classification," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[17] K. Pillutla, K. Malik, A.-R. Mohamed, M. Rabbat, M. Sanjabi, and L. Xiao, "Federated learning with partial model personalization," in *Proc. Int. Conf. Machine Learning (ICML)*, 2022, pp. 17 716–17 758.

[18] S. Lee, T. Zhang, and A. S. Avestimehr, "Layer-wise adaptive model aggregation for scalable federated learning," in *Proc. AAAI Conf. Artificial Intelligence*, 2023, pp. 8491–8499.

[19] H. B. McMahan, D. Ramage, K. Talwar, and L. Zhang, "Learning differentially private recurrent language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.

[20] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Machine Learning and Systems (MLSys)*, 2020, pp. 429–450.

[21] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.

[22] Q. Li, Y. Diao, Q. Chen, and B. He, "Federated learning on non-IID data silos: An experimental study," in *Proc. IEEE Int. Conf. Data Engineering (ICDE)*, 2022, pp. 965–978.

[23] T. Li, S. Hu, A. Beirami, and V. Smith, "Ditto: Fair and robust federated learning through personalization," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 6357–6368.

[24] X. Li, M. Jiang, X. Zhang, M. Kamp, and Q. Dou, "FedBN: Federated learning on non-IID features via local batch normalization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[25] J. Wang and G. Joshi, "Cooperative SGD: A unified framework for the design and analysis of local-update SGD algorithms," *Journal of Machine Learning Research*, vol. 22, no. 213, pp. 1–50, 2021.

[26] A. Reisizadeh, A. Mokhtari, H. Hassani, A. Jadbabaie, and R. Pedarsani, "FedPAQ: A communication-efficient federated learning method with periodic averaging and quantization," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2020, pp. 2021–2031.

[27] X. Ma, J. Zhang, S. Guo, and W. Xu, "Layer-wised model aggregation for personalized federated learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10 092–10 101.

[28] Y. A. U. Rehman, Y. Gao, P. P. B. de Gusmão, M. Alibeigi, J. Shen, and N. D. Lane, "L-DAWA: Layer-wise divergence aware weight aggregation in federated self-supervised visual representation learning," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 16 464–16 473.

[29] H. Wang, H. Xu, Y. Li, Y. Xu, R. Li, and T. Zhang, "FedCDA: Federated learning with cross-rounds divergence-aware aggregation," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.

[30] K. Yang, T. Jiang, Y. Shi, and Z. Ding, "Federated learning via over-the-air computation," *IEEE Transactions on Wireless Communications*, vol. 19, no. 3, pp. 2022–2035, 2020.

[31] K. B. Letaief, W. Chen, Y. Shi, J. Zhang, and Y.-J. A. Zhang, "The roadmap to 6G: AI empowered wireless networks," *IEEE Communications Magazine*, vol. 57, no. 8, pp. 84–90, 2019.

[32] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Network*, vol. 34, no. 3, pp. 134–142, 2020.

[33] I. Mironov, "Rényi differential privacy," in *Proc. IEEE Computer Security Foundations Symposium (CSF)*, 2017, pp. 263–275.

[34] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[35] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients – how easy is it to break privacy in federated learning?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[36] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *Proc. IEEE Computer Security Foundations Symposium (CSF)*, 2018, pp. 268–282.
[37] 3GPP, "Study on scenarios and requirements for next generation access technologies," 3rd Generation Partnership Project, Tech. Rep. TR 38.913, 2022.
[38] I. Mironov, K. Talwar, and L. Zhang, "Rényi differential privacy of the sampled Gaussian mechanism," *arXiv preprint arXiv:1908.10530*, 2019.
[39] B. Strack, J. P. DeShazo, C. Gennings, J. L. Olmo, S. Ventura, K. J. Cios, and J. N. Clore, "Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records," *BioMed Research International*, vol. 2014, p. 781670, 2014.
[40] R. Detrano, A. Janosi, W. Steinbrunn, M. Pfisterer, J.-J. Schmid, S. Sandhu, K. H. Guppy, S. Lee, and V. Froelicher, "International application of a new probability algorithm for the diagnosis of coronary artery disease," *The American Journal of Cardiology*, vol. 64, no. 5, pp. 304–310, 1989.
[41] J. O. du Terrail, S.-S. Ayed, E. Cyffers *et al.*, "FLamby: Datasets and benchmarks for cross-silo federated learning in realistic healthcare settings," in *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2022.
[42] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi, "The eICU collaborative research database, a freely available multi-center database for critical care research," *Scientific Data*, vol. 5, p. 180178, 2018.
[43] A. Johnson, T. Pollard, O. Badawi, and J. Raffa, "eICU collaborative research database demo (version 2.0.1)," PhysioNet, 2021, https://physionet.org/content/eicu-crd-demo/2.0.1/.
[44] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, p. 41, 2023.
[45] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting deep learning models for tabular data," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, pp. 18 932–18 943.